Binomial Distribution

1. The probability that a patient recovers from a rare blood disease is 0.4. If 15 people are known to have contacted this disease, what is the probability that
a. atleast 10 survive b. from 3 to 8 survive
c. Exactly 5 survive.

Ans. Given

$n = 15$

$p = 0.4$

$q = 1 - p = 1 - 0.4 = 0.6$

$$nC_r = \frac{n!}{r!(n-r)!}$$

$$P(X = K) = \binom{n}{K} p^K q^{n-K}$$

a. $$P(X \geq 10) = \sum_{K=10}^{15} \binom{15}{K} (0.4)^K (0.6)^{15-K}$$

i.e $P(X \geq 10) = P(X=10) + P(X=11) + P(X=12) + P(X=13) + P(X=14) + P(X=15)$

$$P(X=10) = {}^nC_r\, p^q\, n^{n-r}$$

$$= {}^{15}C_{10}\, p^{10} q^{15-10}$$

$$= \frac{15!}{10!(15-10)!} p^{10} q^{15-10}$$

$$= \frac{15!}{10!\,5!} (0.4)^{10} (0.6)^5$$

$$= 0.0052$$

b. $P(3 \leq X \leq 8) = \sum_{k=3}^{8} \binom{15}{k} (0.4)^k (0.6)^{15-k}$

$= P(X=3) + P(X=4) + P(X=5) + P(X=6) + P(X=7) + P(X=8)$

$= 0.6654$

c. $P(X=5) = \binom{15}{5} (0.4)^5 (0.6)^{15-5}$

$= \frac{15!}{5!(15-5)!} (0.4)^5 (0.6)^{10}$

$= 3003 \times 0.01024 \times 0.00604$

$= 0.1859.$

2. Suppose that for a very large shipment of integrated circuit chips, the probability of failure for any chip is 0.10. Assuming that the assumptions underlying the binomial distributions are met. Find the probability that at most 3 chips fail in a random sample of 20.

Given

$n = 20$

$p = 0.10$

$q = 1-p = 1-0.10 = 0.9$

$P(X \leq 3) = \sum_{k=0}^{3} \binom{20}{k} p^k q^{20-k}$

$= \sum_{k=0}^{3} \binom{20}{k} (0.10)^k (0.90)^{20-k}$

$P(X=0) = {}^{n}C_r p^r q^{n-r} = {}^{20}C_0 (0.1)^0 (0.9)^{20} = 0.1216$

$P(X=1) = {}^{20}C_1 (0.1)^1 (0.9)^{19} = \frac{20!}{1!(19)!} \cdot 0.1 \cdot (0.9)^{19}$

$= 0.2702$

Similarly calculate for $P(X=2)$ & $P(X=3)$

$P(X=0)$

$P(X \leq 3) = P(X=1) + P(X=2) + P(X=3)$

$= 0.1216 + 0.2702 + 0.2852 + 0.1893$

≈ 0.8663

Exercises

1) Fair coin is tossed 6 times. find the probability of getting a) Exactly 2 heads b) at least four heads)
Hind $P(X=2)$, $P(X \geq 4)$

2) A safty engineer claims that only 40% of all workers wear saftey helmets when they eat lunch at the workplace. Assume that claim is right. find the probability that ~~out a~~ 4 out of 6 workers will be wearing helmets while having lunch at the workplace

Poisson Distribution.

1. $$P(X=x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

During laboratory experiment, the average number of radioactive particles passing through a counter in 1 millisecond is 4. What is the probability that 6 particles enter the counter in a given millisecond.

Given

$\lambda = 4, \quad x = 6$

$$P(X=6) = \frac{4^6 \cdot e^{-4}}{6!} = 0.1042$$

2. The number of customers arriving per hour at a certain automobile service facility is assumed to follow a poisson distribution with mean $\lambda = 7$

a. Compute the probability that more than 10 customers will arrive in a 2 hour period

b. What is the mean number of interval during a 2 hour period.

Given

$\lambda = 7$, for 2hr arrival rate $= 2 \times 7 = 14$

a. $P(X > 10) = 1 - P(X \leq 10)$

$$= 1 - \sum_{K=0}^{10} \frac{14^K e^{-14}}{K!}$$

b. Mean no. of arrival during 2 hours

$\mu = \lambda_{2h} = 2 \times 7 = 14$

8. A certain area of the eastern United States is on average, hit by 6 hurricanes a year. find the probability that a given year that area will be hit by

a. fewer than 4 hurricanes

b. anywhere from 6 to 8 hurricanes

Given

$\lambda = 6$

$P(X=K) = \frac{\lambda^K e^{-\lambda}}{K!}$

$P(X<4) = P(X \leq 3) = \sum_{K=0}^{3} \frac{e^{-6} 6^K}{K!}$

$P(X=0) = \frac{6^0 \cdot e^{-6}}{0!} = 0.00248$

$P(X=1) = \frac{6^1 \cdot e^{-6}}{1!} = 0.01487$

$P(X=2) = 0.4462$

$P(X=3) = 0.08924$

$P(X<4) = P(X=0) + P(X=1) + P(X=2) + P(X=3)$

$= 0.00248 + 0.01487 + 0.4462 + 0.08924$

≈ 0.1512

b. b. $P(6 \leq X \leq 8)$

$= P(X=6) + P(X=7) + P(X=8)$

$P(X=6) = \frac{e^{-6} \cdot 6^6}{6!} = 0.1606$

$P(X=7) = \frac{e^{-6} 6^7}{7!} = 0.1374$

$P(X=8) = \frac{e^{-6} \cdot 6^8}{8!} = 0.1030$

Exercise

1) Determine the probability p that there are 3 defective items in a sample of 100 items if 2% made in factory are defective.

2. On average, 3 traffic accidents per month occur at a certain intersection. What is the probability that in any given month at this intersection.

a. Exactly 5 accident will occur?

b. at least 2 accident will occur?

Hint

$P(X=5)$, & $P(X \geq 2) = 1 - P(X < 2)$

3. On average, a textbook author makes two word processing errors per page on first draft of textbook. What is the probability that on next page, author will make

a) 4 or more errors

b) no errors.

Normal Distribution

1. A certain battery last on average 3.0 years with a standard deviation of 0.5 yrs. Assume that battery life is normally distributed. find the probability that given battery will last less than 2.3 yrs.

Given

$x = 2.3$

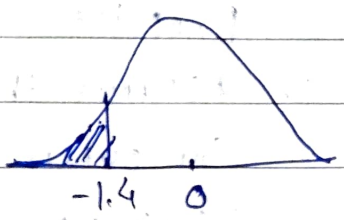
$\bar{x} = 3$

$\sigma = 0.5$

$$Z = \frac{x - \bar{x}}{\sigma} = \frac{2.3 - 3}{0.5} = -1.4$$

$P(x < 2.3)$

$= P(Z < -1.4)$



from Z table

$= 0.4192$

$\therefore P(Z \leq -1.4) = 0.5 - 0.4192 = 0.0808$

2. An electrical ~~fan~~ firm manufacturers light bulbs that have a life before burn-out that is normally distributed with mean 800 hrs and S.D. of 40 hrs. find the probability that a bulb life ~~is in between~~ between 778 and 834 hrs.

Given

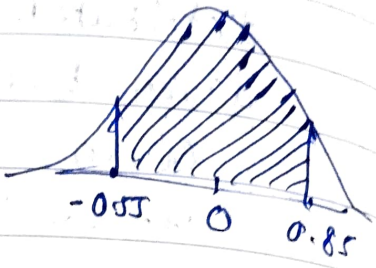
$\bar{x} = 800$, $\sigma = 40$

$P(778 < x < 834) = ?$

$$Z_1 = \frac{x - \bar{x}}{\sigma} = \frac{778 - 800}{40} = -0.55$$

$$Z_2 = \frac{x - \bar{x}}{\sigma} = \frac{834 - 800}{40} = 0.85$$

$P(-0.55 < X < 834)$



~~$P(-0.55)$~~

Area(0 to -0.55) + Area(0 to 0.85)

$= 0.2088 + 0.3023 = 0.5111$

Exercise

1. Assume that average life span of computers produced by a company is 2040 hrs with S.D of 60 hrs. Find expected number of computers whose life span is
   a. more than 2150 hrs
   b. less than 1950 hrs
   c. more than 1920 hrs & less than 2160 hrs
2. The weights of a large number of miniature poodles are approximately normally distributed with a mean of 8 kg and S.D of 0.9 kg. find fraction of these poodles with weights
   a. over 9.5 kg
   b. between 7.3 and 9.1 kgs inclusive

t Distribution

$$t = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \qquad \boxed{df = n-1}$$

1. A chemical engineer claims that population mean yield of certain batch process is 500 grams per milliliter of raw material. To check claim he samples 25 batches each month. If t value falls between $-t_{0.05}$ and $t_{0.05}$, he satisfied with claim. What inclusion should he draw from sample that has mean $\bar{x} = 518$ grams per milliliter & sample S.D $s = 40$ grams. Assume distribution is normal.

Given $\bar{x} = 518$, $\mu = 500$, $\sigma = 40$, $n = 25$

$$t = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$= \frac{518 - 500}{40/\sqrt{25}} = 2.25$$

from t table $t_{0.05} = 1.711$ for degree of freedom 24 (25-1)

$2.25 > 1.71$.

The probability of obtaining t value with $v = 24$ is greather than 2.25 is 0.02 approx so Engineer conclude that process produce better product than he thought.

2. A normal population with unknown varian... has a mean of 20. Is one likely to obta... a random sample of size 9 from popula... with mean of 24 and S.D of 4.1 ? If not what conclusion would you draw?

Given

Population of mean - $\mu = 20$

sample mean $\bar{x} = 24$

S (Standard deviation) $= 4.1$

$n = 9$

$df = 9 - 1 = 8$

$$t = \frac{\bar{x} - \mu}{S/\sqrt{n}} = \frac{24 - 20}{4.1/\sqrt{9}} = \frac{4}{4.1/3} = 2.92$$

Central limit theorem.

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

is the standard normal distribution $n(0,1)$ - Mean 0 & variance 1.

1. An electrical firm manufactures light bulb that have a length of life is normally distributed. with mean 800 hrs & S.D of 40 hrs. find probability of random sample of 16 bulbs will have an average life less than 775 hrs.

Given

$\mu_{\bar{x}}$ - sample distribution of mean = 800

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{40}{\sqrt{16}} = \frac{40}{4} = 10$$

$\bar{x} = 775$

$$z = \frac{775 - 800}{10} = -2.5$$

$P(x < 775) = P(Z < -2.5) = 0.0062$

Exercise

1. The height of 1000 students are approx normally distributed with a mean of 174.5 cm & S.D of 6.9 cm. Suppose 200 samples of size 25 are drawn from population find -

i. Mean & S.D. of sampling distribution of $\bar{x}$
ii. Number of sample means fall between 172.5 and 175.8 cm
iii. Number of sample means falling below 172.0 cm.

F Distribution

$$F = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} = \frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2}$$

with $v_1 = n_1 - 1$ & $v_2 = n_2 - 1$ degree of freedom.

1. The household net expenditure on heath care in south & north India in two sample

south 15.0 8.0 3.8 6.4 27.4 19.0 35.3 13.6
north 18.8 23.1 10.3 8.0 18.0 10.2 15.2 19.0 20.2

Test equality of variance

$$F = \frac{S_1^2}{S_2^2}$$

$$S_1^2 = \frac{\sum (x - \bar{x})^2}{n_1 - 1}$$

$$\bar{x}_1 = \sum_{i=1}^{8} x_{1i} = 15 + 8 + 3.8 + 6.4 + 27.4 + 19.0 + 35.3 + 13.6 = 128.5$$

$$S_1^2 = \frac{1}{n_1 - 1}\left[\sum_{i=1}^{8} x_{1i}^2 - \frac{(\sum x_{1i})^2}{n_1}\right]$$

$$s_1^2 = \frac{1}{8-1}\left[2887.21 - \frac{(128.5)^2}{8}\right] = 117.596$$

$$s_2^2 = \bar{x}_2 = \sum_{i=1}^{9} x_{2i} = 18.8 + 23.1 + 10.3 + 8.0 + 18.8 + 10.2 + 15.2 + 19.0 + 20.2 = 142.8$$

$$s_2^2 = \frac{1}{9-1}\left[2485.26 - \frac{(142.8)^2}{9}\right] = 27.4375$$

$$F = \frac{s_1^2}{s_2^2} = \frac{117.596}{27.4375} = 4.28$$

$4.28 > 1$, so variances are significantly different (4.28 is far away from 1)

Exercise

1. Can we conclude that two population variances are equal for following data of state & private university.

State	8350	8260	8130	8340	8070	
private	7890	8140	7900	7950	7840	7920

Bayes Theorem

$$P(E_1|A) = \frac{P(E_1) \cdot P(A|E_1)}{P(A)}$$

1. An insurance company insured 1000 scooter drivers, 3000 car drivers & 6000 truck drivers. The probability that scooter, car & truck drivers meet an accident are 0.02, 0.04, 0.25 respectively. One of the insured person meets with an accident. What is the probability that he is a
a) car driver
b) Truck driver

Let S, C & T be events that insured person is scooter driver, car driver & truck driver.

Total no. of drivers = 1000 + 3000 + 6000 = 10000

$P(C) = \frac{3000}{10000} = 0.3$

$P(S) = \frac{1000}{10000} = 0.1$

$P(T) = \frac{6000}{10000} = 0.6$

Given

$P(A|S) = 0.02$

$P(A|C) = 0.04$

$P(A|T) = 0.25$

Total Probability = $P(A)$ =

$P(S) \cdot P(A|S) + P(C) \cdot P(A|C) + P(T) \cdot P(A|T)$

$= 0.1 \times 0.02 + 0.3 \times 0.04 + 0.6 \times 0.25$

$= 0.164$

i. Probability that person is car driver

$$P(C|A) = \frac{P(C) \cdot P(A|C)}{P(A)}$$

$$= \frac{0.3 \times 0.04}{0.164} = 0.073$$

ii. Probability that person is truck driver

$$P(T|A) = \frac{P(T) \cdot P(A|T)}{P(A)}$$

$$= \frac{0.6 \cdot 0.25}{0.164} = 0.91$$

Exercise

Companies B1, B2, B3 produce 30%, 45% & 25% of the cars respectively. It is known that 2%, 3% & 2% of the cars produced are defective.

1) What is the probability that a car purchased is defective

2) If a car purchased is found to be defective What is probability that this car is of company B3.

Solve the linear equation by Gaus Elimination

$2x - 2y + 3z = 2$

$x + 2y - z = 3$

$3x - y + 2z = 1$

Augmented Matrix

$$\left[\begin{array}{ccc|c} 2 & -2 & 3 & 2 \\ 1 & 2 & -1 & 3 \\ 3 & -1 & 2 & 1 \end{array}\right]$$

$R_1 \leftrightarrow R_2$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 2 & -2 & 3 & 2 \\ 3 & -1 & 2 & 1 \end{array}\right]$$

$R_2 = R_2 - 2R_1$

$R_3 = R_3 - 3R_1$

for $-2 = -2 - 2(2) = -6$

$3 = 3 - 2(-1) = 5$

$-1 = -1 - 3(2) = -7$

$2 = 2 - 3(-1) = 5$

for $2 = 2 - 2(3) = -4$

$1 = 1 - 3(3) = -8$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -6 & 5 & -4 \\ 0 & -7 & 5 & -8 \end{array}\right]$$

$R_3 = R_3 - \frac{7}{6} R_2$

for $-7 = -7 - \frac{7}{6} \times -6 = 0$

for $5 = 5 - \frac{7}{6} \times 5 = \frac{-5}{6}$

for $-8 = -8 - \frac{7}{6} \times -4 = \frac{-10}{3}$

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & -6 & 5 & -4 \\ 0 & 0 & -5/6 & -10/3 \end{array}\right]$$

$x + 2y - z = 3 \quad -(1)$

$-6y + 5z = -4 \quad -(2)$

$-\frac{5}{6} z = \frac{-10}{3} \quad -(3)$

$$\frac{-5}{6}z = \frac{-10}{3}$$

$$z = \frac{-10^{2}}{3_{1}} \times \frac{6^{2}}{-5_{1}}$$

$$\boxed{z = 4}$$

Substitute z in eq. (2)

$$-6y + 5z = -4$$

$$-6y + 5 \times 4 = -4$$

$$-6y + 20 = -4$$

$$-6y = -4 - 20$$

$$-6y = -24$$

$$\boxed{y = 4}$$

Substitute y & z in eq. (1)

$$x + 2y - z = 3$$

$$x + 2 \times 4 - 4 = 3$$

$$x + 8 - 4 = 3$$

$$x + 4 = 3$$

$$x = 3 - 4$$

$$\boxed{x = -1}$$

Exercise

1. $2x + 4y + 6z = 22$

   $3x + 8y + 5z = 27$

   $x + y + 2z = 2$

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2. $x - 2y + z = 0$
$2x + y - 3z = 5$
$4x - 7y + z = -1$

Eigen values & Eigen vectors

1. find eigen values & eigen vectors of A

$$A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

$$(A - \lambda I) = 0$$

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{bmatrix}$$

$$= (2-\lambda)(2-\lambda) - (-1 * -1) = 0$$
$$= \lambda^2 - 4\lambda + 4 - 1 = 0$$
$$= \lambda^2 - 4\lambda + 3 = 0$$
$$(\lambda - 3)(\lambda - 1) = 0$$
$$\lambda = 3 \text{ or } \lambda = 1$$

for $\lambda = 1$

$$\begin{bmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{bmatrix} = \begin{bmatrix} 2-1 & -1 \\ -1 & 2-1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\therefore x - y = 0$$

$$\frac{x}{1} = \frac{y}{1}$$

$$\text{vector} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

for $\lambda = 3$

$$\begin{bmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{bmatrix} = \begin{bmatrix} 2-3 & -1 \\ -1 & 2-3 \end{bmatrix}$$

$$\begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\therefore -x - y = 0$$

$$-x = y$$

$$\frac{-x}{-1} = \frac{y}{-1}$$

$$\frac{x}{+1} = \frac{y}{-1}$$

$$\text{vector} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

2) $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$

$= \begin{bmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{bmatrix}$

$= 8-\lambda\left[(7-\lambda)(3-\lambda)-(-4\times-4)\right]-(-6)\left[-6(3-\lambda)-(-4\times2)\right]$
$+2\left[(-6\times-4)-(7-\lambda)2\right]$

$= 8-\lambda\left[\lambda^2-10\lambda+21-16\right]+6\left[-18+6\lambda+8\right]$
$+2\left[24-14+2\lambda\right]$

$= \cancel{8\lambda^2-80\lambda+40}$
$8-\lambda\left[\lambda^2-10\lambda+5\right]+6\left[-10+6\lambda\right]+$
$2\left[10+2\lambda\right]$

$= 8\lambda^2-80\lambda+40-\lambda^3+10\lambda^2-5\lambda-60+36\lambda$
$+20+4\lambda$

$= -\lambda^3+18\lambda^2-45\lambda = 0$

$\lambda^3-18\lambda^2+45\lambda = 0$

Put $\lambda = 3$

$$A = 3\begin{vmatrix} 1 & -18 & 45 \\ 0 & 3 & -45 \\ 1 & -15 & 0 \end{vmatrix}$$

So roots are

$(\lambda-3)(\lambda^2-15\lambda) = 0$

$(\lambda-3)(\lambda)(\lambda-15) = 0$

$$\boxed{\lambda = 0,\ \lambda = 3,\ \lambda = 15}$$

for $\lambda = 0$ (Cramer's Rule)

Consider first 2 equation

$$\begin{bmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{bmatrix}$$

$$= \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$8x - 6y + 2z = 0$ — (1)

$-6x + 7y - 4z = 0$ — (2)

$2x - 4y + 3z = 0$ — (3)

Consider first 2 equations

$$\frac{x}{\begin{vmatrix} -6 & 2 \\ 7 & -4 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} 8 & 2 \\ -6 & -4 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 8 & -6 \\ -6 & 7 \end{vmatrix}}$$

$$\frac{x}{24-14} = \frac{-y}{-32+12} = \frac{z}{56-36}$$

$$\frac{x}{10} = \frac{-y}{-20} = \frac{z}{20}$$

$$\frac{x}{1} = \frac{y}{2} = \frac{z}{2}$$

vector $\begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$

for $\lambda = 3$

$$\begin{bmatrix} 8-3 & -6 & 2 \\ -6 & 7-3 & -4 \\ 2 & -4 & 3-3 \end{bmatrix}$$

$$\begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$5x - 6y + 2z = 0 \quad \text{---(1)}$

$-6x + 4y - 4z = 0 \quad \text{---(2)}$

$$\frac{x}{\begin{vmatrix} -6 & 2 \\ 4 & -4 \end{vmatrix}} = \frac{-y}{\begin{vmatrix} 5 & 2 \\ -6 & -4 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 5 & -6 \\ -6 & 4 \end{vmatrix}}$$

$$\frac{x}{16} = \frac{-y}{-8} = \frac{z}{-16}$$

$$\frac{x}{2} = \frac{y}{1} = \frac{z}{-2}$$

Vector $= \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$

for $\lambda = 15$
do same procedure

vector $= \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$

Digonalization

$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$, Verify $P^{-1} A P = D$

$|A - \lambda I| = \begin{vmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{vmatrix} = 0$

$(4-\lambda)(3-\lambda) - 2 = 0$
$\lambda^2 - 7\lambda + 12 - 2 = 0$
$\lambda^2 - 7\lambda + 10 = 0$
$(\lambda - 5)(\lambda - 2) = 0$
$\lambda = 5, \quad \lambda = 2$

for $\lambda = 2$

$$\begin{bmatrix} 4-2 & 1 \\ 2 & 3-2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$2x + y = 0$

$$2x = -y$$

$$\frac{2x}{2} = \frac{-y}{2}$$

$$\frac{x}{1} = \frac{y}{-2}$$

$$X_1 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

for $\lambda = 5$

$$\begin{bmatrix} 4-5 & 1 \\ 2 & 3-5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$= -x + y = 0$$

$$-x = -y$$

$$\frac{x}{1} = \frac{y}{1}$$

$$X_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$P = \begin{bmatrix} X_1 & X_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix}$$

$$P^{-1} = \frac{1}{3} \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$$

$$P^{-1}AP = \frac{1}{3}\begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 2 & -2 \\ 10 & 5 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 6 & 0 \\ 0 & 15 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$$

Find A^6

$$A^6 = PD^6P^{-1}$$

$$= \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix}\begin{bmatrix} 2^6 & 0 \\ 0 & 5^6 \end{bmatrix}\frac{1}{3}\begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 64 & 15625 \\ -128 & 15625 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 31314 & 15561 \\ 31122 & 15753 \end{bmatrix}$$

$$A^6 = \begin{bmatrix} 10438 & 5187 \\ 10374 & 5251 \end{bmatrix}$$